A COMPARISON OF MONTE CARLO SIMULATION TECHNIQUES FOR COMPOSITE POWER SYSTEM RELIABILITY ASSESSMENT

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Abstract - The paper illustrates three Monte Carlo simulation approaches and compares the sequential technique with non-sequential methods such as state sampling and state transition sampling, when applied to composite power system reliability assessment. The three different Monte Carlo simulation methods have been applied to a composite generation and transmission test system utilizing an annualized peak load and the results are compared. The paper also illustrates the utilization of an annual chronological load curve for each load bus in the test system and a sequential Monte Carlo simulation approach for composite system reliability assessment. An equivalent method, using a load duration curve of the system load and an enumeration process have been applied to the load model and the results are compared in this paper.

1. INTRODUCTION

Composite generation and transmission system reliability evaluation [1] is of considerable interest to power system engineers and managers. There are two fundamental approaches to composite system reliability evaluation - analytical enumeration and Monte Carlo simulation. Monte Carlo methods are generally more flexible when complex operating conditions and system considerations (such as multi-derated states, chronology, reservoir operating rules, bus load uncertainty, weather effects) need to be incorporated.

There are two basic techniques utilized in Monte Carlo applications to power system reliability evaluation. These are known as the sequential and non-sequential techniques. In the non-sequential method, the states of all components are sampled and a non-chronological system state is obtained. In the sequential approach, the up and down cycles of all components are simulated and a system operating cycle is obtained by combining all the component cycles. The sequential technique permits chronological issues to be considered and distributions of the reliability indices to be calculated. This method usually requires a larger investment in computing time and effort compared to the non-sequential technique.

Several models [2 - 8] have been implemented to perform composite system analysis using Monte Carlo simulation. The sequential simulation approach is the most fundamental technique available and can be used to provide a wide range of indices. It can also be used to provide estimates which can serve as benchmarks against which other approximate techniques can be compared. This paper illustrates three Monte Carlo simulation techniques and compares the sequential method with non-sequential approaches.

2.0 STATE SAMPLING TECHNIQUE

In the state sampling (non-sequential) method, the states of all components are sampled and a non-chronological system state is obtained. The basic sampling procedure is conducted by assuming that the behavior of each component can be categorized by a uniform distribution [0,1]. In the case of a two state component representation the probability of outage is given by the component forced outage rate.

In order to illustrate this, let the state of a system be represented by a vector $S = (S_1, S_2, ..., S_i, ..., St)$, where S_i denotes the state of the ith component. The set S of t components includes the state of each element of the system (generators, lines, transformers, etc.). Let FOR_i be component i 's forced outage rate. The state S_i of the ith component is determined using the linear congruential method to draw a pseudo-random number x distributed uniformly under [0,1].

$$S_i = 0 \text{ (Normal state)} \quad x >= FOR_i \qquad (1)$$

$$S_i = 1 \text{ (Outage state)} \quad 0 <= x < FOR_i \qquad (2)$$

It is possible to calculate the associated probability of occurrence for every system state S [3, 8]. The steps followed in using the state sampling technique for composite system adequacy assessment can be briefly summarized as follows:

1. A system state is sampled using the state sampling technique.
2. If this state is a normal state , then no load curtailment exists. If this is a contingency state, load curtailment may be required. The transmission line power flows are calculated using DC load flow.
3. The linear programming minimization model [8] is solved to reschedule generation, alleviate line overloads and avoid load curtailment if possible or to minimize total load curtailment if unavoidable.
4. The adequacy indices [9] are accumulated and Steps 1 - 3 are repeated until the coefficient of variation of the Expected Demand Not Served (EDNS) [3] is less than the tolerance error.

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The state sampling method is relatively simple and it is only necessary to generate uniformly distributed random numbers between [0,1]. It does not require sampling from distribution functions as in a sequential approach. The state sampling approach, without the use of additional enumeration methods estimates the frequency of load curtailments as the sum of the occurrences of load curtailment states. This is only an upper bound on the frequency index. The state transition sampling technique provides an estimate of the actual frequency index.

2.1 STATE TRANSITION SAMPLING TECHNIQUE

The state transition sampling technique [10, 11] focuses on state transitions of the whole system rather than on component states or component state durations. This method does not require sampling state duration distribution functions of all the components and the storage of chronological information as required in the sequential method. It is therefore, computationally faster than the sequential approach, but slower than the state sampling method. An important restriction in this method is the assumption of exponential distributions for all state residence times. The steps followed in composite system adequacy assessment are briefly described below.

1. The simulation process starts from the normal system state in which all the generating units and transmission lines are in the up state.
2. If the present system state is a contingency state in which at least one component is in the outage state, the minimization model of load curtailment is used to evaluate the adequacy for this system state. Otherwise, it is possible to proceed to the next step without utilizing the minimization model.
3. A uniform distribution random number is generated to determine the next system state using the state transition sampling procedure [10].
4. The process is repeated from Step 2 until the convergence criterion is satisfied.

2.2 SEQUENTIAL METHOD

The sequential or state duration sampling method is based on sampling the probability distribution of the component state duration. The sequential Monte Carlo approach can be used to model all contingencies and operating characteristics inherent in the system. It can be easily used to calculate the actual frequency index and any state duration distribution function can be considered. The major steps in using the sequential procedure for composite system reliability assessment, are as follows:

1. The times to failure and times to repair of all components (generating units, transmission lines) for a yearly sequence are produced by sampling the appropriate probability distributions. In this procedure, the state residence times are assumed to be exponentially distributed. A random variate T with an underlying exponential distribution has a probability density function (p.d.f) [12, 13]:

$$f_T(t) = \lambda e^{-\lambda t} \qquad (3)$$

where λ is the mean value of the distribution. Using the inverse transform method [13], the random variate T is given by:

$$T = -\ln(1-U) / \lambda \qquad (4)$$

where U is a uniformly distributed random number over the interval (0,1). The availability of all system components are obtained for the simulated hour.
2. The chronological hourly load model, if available for each load bus, gives the load for the simulated hour at each bus.
3. The simulated operation of the system is assessed using the linear programming minimization model and the adequacy indices are accumulated.
4. Steps 2 - 3 are performed for the yearly sequence of system states. The yearly adequacy indices are accumulated as $F(X_j)$ where X_j is the sequence of system states in year j and $F(X_j)$ is the reliability index function over the year j. In order to evaluate the LOLP for instance, $F(X_j)$ is the sum of durations of all failure states divided by 8736 in the year j.
5. If the coefficient of variation of the chosen index [14] is greater than the tolerance level, steps 1 - 4 are repeated until convergence is achieved.

This method when compared to the other two approaches requires more computing time and storage since it is necessary to generate a random variate following a given distribution for each component and to store information on chronological component state transition processes of all the components in a long time span [15].

3.0 APPLICATION TO A TEST SYSTEM UTILIZING AN ANNUALIZED LOAD

These three Monte Carlo simulation methods have been applied to the Roy Billinton test system (RBTS) [16] and the reliability indices estimated for an annualized peak load of 185 MW. The reliability indices estimated are the probability of load curtailed (PLC), expected frequency of load curtailments (EFLC) and expected energy not served (EENS) [9]. Table 1 provides the load bus indices for the test system, obtained using the three simulation methods. Table 2 describes some of the major system indices estimated using the three methods.

The sequential simulation process was continued until the coefficient of variation of the expected demand not served (EDNS) reached 6%.

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Table 1: Load Bus Indices for The Test System using The Three Simulation Methods

Bus	PLC			EFLC (/ Yr.)			EENS (Mw h / Yr.)		
No:	Seq.	SS	STS	Seq.	SS	STS	Seq.	SS	STS
2	0.00109	0.00128	0.00122	0.453	0.625	0.515	39.84	47.17	44.08
3	0.00397	0.00400	0.00421	1.72	1.845	1.784	482.26	482.65	505.93
4	0.00662	0.00644	0.00637	2.30	2.884	2.201	357.57	334.59	345.17
5	0.00019	0.00022	0.00022	0.23	0.280	0.183	3.851	7.163	4.636
6	0.00113	0.00124	0.00112	0.88	1.244	0.979	197.49	211.06	191.64

Table 2: System Reliability Indices for An Annualized Peak Load of 185 MW

System Reliability Indices	Sequential	State Transition Sampling	State Sampling
PLC	0.00989	0.00985	0.01014
EFLC (/Yr)	4.13	4.14	5.21
EENS (Mwh / Yr.)	1081.01	1091.46	1082.63
Simulation period	150 years	50000 samples	50000 samples
CPU (sec)	733	62	36

The frequency obtained using the state sampling method is higher, as expected. The studies were conducted on a SUN station. The state sampling technique requires less memory and CPU time compared to the other two methods, but has the disadvantage of being incapable of directly providing frequency and duration related indices.

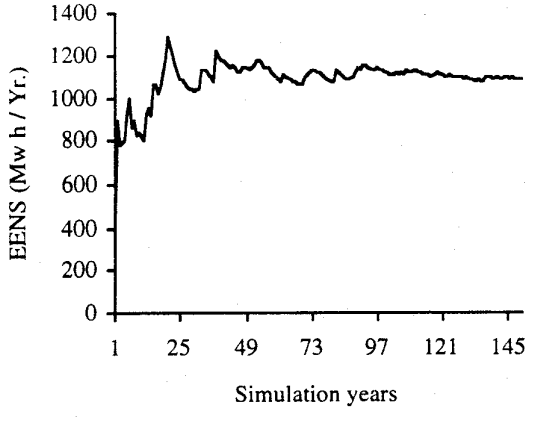


Fig. 1: EENS vs. simulation years

Figs. 1 and 2 shows the system EENS and system EFLC versus the simulation years for the sequential method.

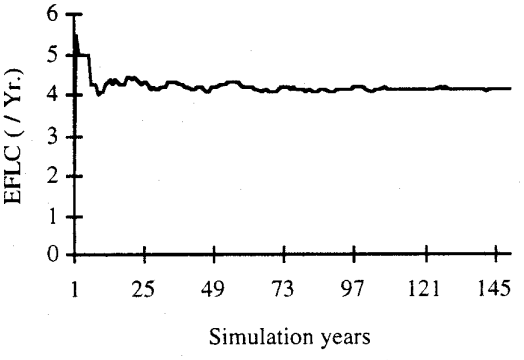


Fig. 2: EFLC vs. simulation years

4.0 SEQUENTIAL SIMULATION APPLIED TO TIME VARYING LOADS AT LOAD BUSES

Earlier studies [7,12] have used the IEEE Reliability Test System (IEEE - RTS) [17] hourly load model for the overall system. The load model information provided, can be used to calculate system hourly loads for one year on a per unit basis, expressed in a chronological fashion so that daily, weekly and seasonal patterns can be developed. The load model is sufficient for generating capacity reliability studies such as loss of load or energy expectation analysis. The published information, however, is not adequate for composite system studies since the IEEE-RTS load data is specified as total system demand and does not indicate how individual bus loads vary during the period concerned.

A more comprehensive load model would recognize that individual buses have different load curves which depend on the mix of customer classes at that bus. Different hourly load curves at each bus can be developed but collecting this data is, however, a difficult task and it is therefore, not

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generally available. This necessitated the development of a load model using a bottom - up approach starting from the customer sectors present at each bus.

Chronological load curves for all load buses of the RBTS have been developed, considering the different mix of customer classes at each bus [14]. The load curves for the customer sectors were developed taking into account daily, weekly and seasonal patterns. The customer composition at each load bus for the RBTS is shown in Fig. 3.

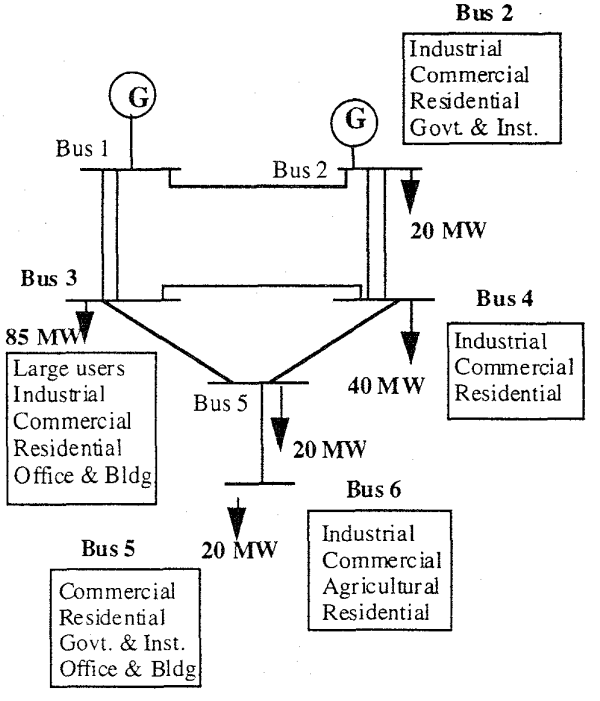


Fig. 3: Single line diagram of the RBTS with customer composition

Figs. 4 and 5 show the load curves for bus 6 and bus 3 of the test system for the first day of the year.

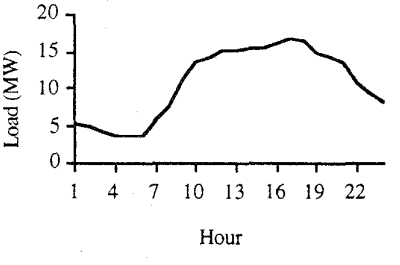


Fig. 4: Load Curve for Bus 6 of the Test System

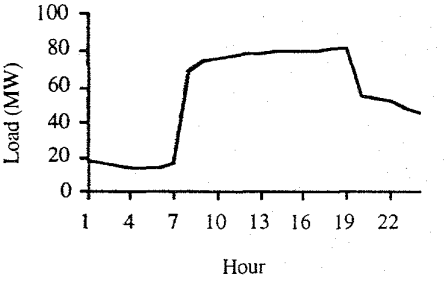


Fig. 5: Load curve for Bus 3 of the Test System

Table 3 presents the system and load point indices after simulating the operation of the system for 1000 years at which point the coefficient of variation for the expected demand not served reaches 6%.

Table 3: Load Bus Indices and System Indices for The Test System

Bus No:	PLC	EFLC (/Yr)	EENS (Mwh/ Yr)
2	0.000045	0.089	1.098
3	0.000287	0.641	19.05
4	0.000290	0.497	14.96
5	0.000010	0.024	0.274
6	0.001180	0.933	111.7

System Indices:

Probability of load curtailment = 0.001599
Expected energy not supplied (Mwh/yr) = 147.113
Expected frequency of load curtailments (/Yr) = 1.700
Average duration of load curtailment (Hrs /dist) = 7.777

Fig. 6 shows the system EENS versus the simulation years.

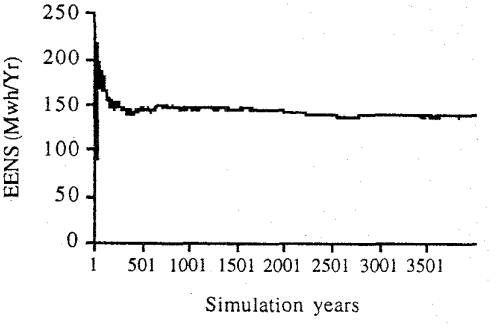


Fig. 6: EENS vs. simulation years

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5.0 EFFECT OF ANALYZING THE SYSTEM USING A LOAD DURATION CURVE

The chronological load model consisting of 8736 points has been translated into the load duration curve shown in Fig. 7.

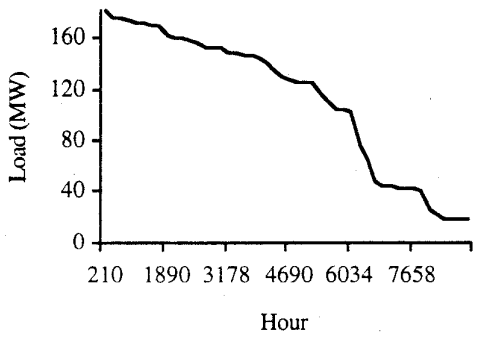


Fig. 7: Load Duration Curve for the Test System

This section discusses the effect of analyzing the system by dividing the load duration curve into a finite number of steps [14]. In the first case, the load duration curve is divided into 10 uniform steps with a step load increment of 10%. Both the state sampling (SS) [8] and the state transition sampling (STS) [10] technique have been applied to this model and the system indices are presented in Table 4.

Table 4: Comparison of System Indices Using The Two Methods - 10 Uniform Load Steps

System Indices	SS	STS
Probability of load curtailed	0.00215	0.00209
Expected energy not supplied (Mwh/ yr)	212.17	202.81
Expected number of load curtailments (/Yr)	1.643	1.288
Average duration of load curtailment (Hrs / disturbance)	11.45	14.23

In the second case, the load duration curve is divided into 15 non-uniform steps (1.0, 0.98, 0.96, 0.94, 0.92, 0.90, 0.86, 0.82, 0.78, 0.74, 0.7, 0.6, 0.5, 0.4, 0.3). The higher load levels have been divided into very fine steps while the lower load levels have a step load decrement of 10%. Table 5 shows the system indices and it can be seen that dividing the load duration curve into non-uniform steps improves the process of obtaining estimates close to those obtained by the sequential method.

Table 5: Comparison of System Indices Using The Two Methods - 15 Non-Uniform Load Steps

System Indices	SS	STS
Probability of load curtailed	0.00184	0.00176
Expected energy not supplied (Mwh/yr)	172.96	161.38
Expected number of load curtailments (/Yr)	1.499	1.163
Average duration of load curtailment (Hrs /disturbance)	10.72	13.18

A 10 step or 15 non - uniform step load model when used with the state sampling or state transition sampling technique over-estimates the reliability indices, other than the frequency. The frequency obtained using the sequential simulation method is higher since the load model transitions are included.

In the third case, the load duration curve is divided into 70 steps with a step load increment of 1%. Tables 6 and 7 present the load bus and system indices using the 70-step load model utilizing the state sampling and the state transition sampling techniques respectively.

Table 6: Load Bus Indices Using The State Sampling (SS) and State Transition Sampling (STS) Methods - 70 Uniform Load Steps

Bus No	PLC		ENLC (/Yr)		EENS (Mwh/ Yr)	
	SS	STS	SS	STS	SS	STS
2	6.5e-05	4.5e-05	0.0391	0.0224	1.78	1.25
3	3.3e-04	2.8e-04	0.1645	0.1186	25.66	20.07
4	2.8e-04	2.7e-04	1.1410	0.1141	16.50	15.19
5	5.0e-06	7.0e-06	0.0069	0.0093	0.159	0.353
6	1.2e-03	1.1e-03	1.1971	0.9522	116.9	109.5

Table 7: System Indices Using The Sequential, State Sampling(SS) and State Transition Sampling (STS) Methods - 70 Uniform Load Steps

System Indices	Sequential	SS	STS
Probability of load curtailed	0.00160	0.00165	0.00153
Expected energy not supplied (Mwh/yr)	147.11	161.03	146.37
Expected number of load curtailments (/Yr)	1.700	1.415	1.115
Average duration of load curtailment (Hrs /disturbance)	7.78	10.19	12.00

Fig. 8 gives the % difference for the EENS obtained using the two non-sequential methods compared with the sequential method.

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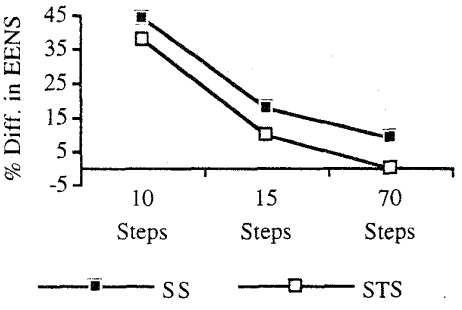


Fig. 8: Comparison of the EENS

The state transition sampling results tend to quickly approach the sequential estimates. The load duration curve, when divided into 70 steps produces estimates of the probability of load curtailed and EENS which are quite close to those obtained by the sequential method.

The sequential method when compared with non-sequential techniques requires a larger computational effort and requires chronological hourly load data at each bus, which may not be easily available. It can be seen, however, that if load duration curve data are available, they can be used for composite system analysis together with a non-sequential approach to obtain reasonably accurate indices.

6.0 CONCLUSIONS

The sequential, state sampling and state transition sampling methods can be effectively used to assess the reliability of composite systems. The sequential simulation technique can be used to provide a comprehensive set of reliability indices including the frequency and duration of interruptions. The effect of using load duration curves and multi-step load models have been presented with reference to a small but practical test system.

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